

- Q.1 Given two independent events A, B such that $P(A) = 0.3$, $P(B) = 0.6$. Determine
 (i) $P(A \text{ and } B)$ (ii) $P(A \text{ and not } B)$ (iii) $P(\text{not } A \text{ and } B)$
 (iv) $P(\text{neither } A \text{ nor } B)$ (v) $P(A \text{ or } B)$

[Ans. (i) 0.18, (ii) 0.12, (iii) 0.42, (iv) 0.28, (v) 0.72]

- Q.2 A card is drawn at random from a well shuffled deck of cards. Find the probability that the card is a
 (i) king or a red card (ii) club or a diamond (iii) king or a queen
 (iv) king or an ace (v) spade or a club (vi) neither a heart nor a king.

[Ans. (i) $\frac{7}{13}$, (ii) $\frac{1}{2}$, (iii) $\frac{2}{13}$, (iv) $\frac{2}{13}$, (v) $\frac{1}{2}$, (vi) $\frac{9}{13}$]

- Q.3 A coin is tossed and a die is thrown. Find the probability that the outcome will be a head or a number greater than 4.

[Ans. $\frac{2}{3}$]

- Q.4 Let A and B be events such that $P(\tilde{A}) = 4/5$, $P(B) = 1/3$, $P(A/B) = 1/6$, then
 (a) $P(A \cap B)$; (b) $P(A \cup B)$; (c) $P(B/A)$; (d) Are A and B independent?

[Ans. (a) $1/18$, (b) $43/90$, (c) $5/18$, (d) NO]

[Sol. (a) $P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{1}{6} \Rightarrow P(A \cap B) = \frac{P(B)}{6} = \frac{1}{18}$ Ans.]

(b) $P(A \cup B) = \frac{1}{5} + \frac{1}{3} - \frac{1}{18} = \frac{18+30-5}{90} = \frac{43}{90}$ Ans.

(c) $P(B/A) = \frac{P(B \cap A)}{P(A)} = \frac{1}{18} \cdot \frac{5}{1} = \frac{5}{18}$ Ans.

(d) $P(A) \cdot P(B) = \frac{1}{5} \cdot \frac{1}{3} = \frac{1}{15} \neq P(A \cap B)$. A & B are not independent]

- Q.5 If A and B are two events such that $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{2}$ and $P(A \text{ and } B) = \frac{1}{8}$, find

(i) $P(A \text{ or } B)$, (ii) $P(\text{not } A \text{ and not } B)$ [Ans. (i) $\frac{5}{8}$, (ii) $\frac{3}{8}$]

- Q.6₁₆₈ A 5 digit number is formed by using the digits 0, 1, 2, 3, 4 & 5 without repetition. The probability that the number is divisible by 6 is :

(A) 8 % (B) 17 % (C*) 18 % (D) 36 %

[Hint: Number should be divisible by 2 and 3.

$n(S) = 5 \cdot 5!$; $n(A) : \text{reject '0'} = 2 \cdot 4!$

reject 3, $4! + 2 \cdot 3 \cdot 3!$

Total $n(A) = 3 \cdot 4! + 6 \cdot 3! = 18 \cdot 3!$

$\therefore p = \frac{18 \cdot 3!}{5 \cdot 5!} = 18\%$]

- Q.7₅₄ An experiment results in four possible out comes S_1, S_2, S_3 & S_4 with probabilities p_1, p_2, p_3 & p_4

respectively. Which one of the following probability assignment is possible.

[Assume S_1, S_2, S_3, S_4 are mutually exclusive]

- (A) $p_1 = 0.25, p_2 = 0.35, p_3 = 0.10, p_4 = 0.05$
 (B) $p_1 = 0.40, p_2 = -0.20, p_3 = 0.60, p_4 = 0.20$
 (C) $p_1 = 0.30, p_2 = 0.60, p_3 = 0.10, p_4 = 0.10$
 (D*) $p_1 = 0.20, p_2 = 0.30, p_3 = 0.40, p_4 = 0.10$

Q.8₈₈ In throwing 3 dice, the probability that atleast 2 of the three numbers obtained are same is

- (A) $1/2$ (B) $1/3$ (C*) $4/9$ (D) none

[Hint: $P(E) = 1 - P(\text{all different}) = 1 - (6/6) \cdot (5/6) \cdot (4/6) = 1 - (120/216) = 4/9$]

Q.9₁₇₃ There are 4 defective items in a lot consisting of 10 items. From this lot we select 5 items at random. The probability that there will be 2 defective items among them is

- (A) $\frac{1}{2}$ (B) $\frac{2}{5}$ (C) $\frac{5}{21}$ (D*) $\frac{10}{21}$

[Hint: $10 \begin{cases} 4D \\ 6G \end{cases} \xrightarrow{5} \begin{cases} 2D \\ 3G \end{cases}$

$$p = \frac{{}^4C_2 \cdot {}^6C_3}{{}^{10}C_5} = \frac{10}{21} \Rightarrow (D)]$$

Q.10₁₃₄ From a pack of 52 playing cards, face cards and tens are removed and kept aside then a card is drawn at random from the remaining cards. If

A : The event that the card drawn is an ace

H : The event that the card drawn is a heart

S : The event that the card drawn is a spade

then which of the following holds ?

- (A*) $9 P(A) = 4 P(H)$ (B) $P(S) = 4 P(A \cap H)$
 (C) $3 P(H) = 4 P(A \cup S)$ (D) $P(H) = 12 P(A \cap S)$

[Hint: $52 \xrightarrow[10's]{\text{face cards}} 36 \begin{cases} 9 H \\ 9 S \\ 9 D \\ 9 C \end{cases}$;

$$P(A) = \frac{1}{9} ; P(H) = \frac{1}{4} ; P(S) = \frac{1}{4} ; P(A \cap H) = \frac{1}{36} ; P(A \cap S) = \frac{1}{36} ; P(A \cup S) = \frac{1}{3}]$$

Q.11₂₂₀ 6 married couples are standing in a room. If 4 people are chosen at random, then the chance that exactly one married couple is among the 4 is :

- (A*) $\frac{16}{33}$ (B) $\frac{8}{33}$ (C) $\frac{17}{33}$ (D) $\frac{24}{33}$

[Hint: $n(S) = {}^{12}C_4 = 55 \times 9$
 $n(A) = {}^6C_1 \cdot {}^5C_2 \cdot 2^2 = 6 \times 10 \times 4$

$$P(E) = \frac{6 \times 10 \times 4}{55 \times 9} = \frac{2.2.4}{11.3} = \frac{16}{33} \text{ Ans}]$$

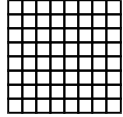
Q.12₄₅ The chance that a 13 card combination from a pack of 52 playing cards is dealt to a player in a game of

bridge, in which 9 cards are of the same suit, is

(A*) $\frac{4 \cdot {}^{13}C_9 \cdot {}^{39}C_4}{{}^{52}C_{13}}$ (B) $\frac{4! \cdot {}^{13}C_9 \cdot {}^{39}C_4}{{}^{52}C_{13}}$ (C) $\frac{{}^{13}C_9 \cdot {}^{39}C_4}{{}^{52}C_{13}}$ (D) none

Q.13₆₄ If two of the 64 squares are chosen at random on a chess board, the probability that they have a side in common is:

- (A) 1/9 (B*) 1/18 (C) 2/7 (D) none



[Hint: $n(S) = {}^{64}C_2 \cdot 2$; $n(A) = \frac{4 \cdot 2 + 6 \cdot 4 \cdot 3 + 36 \cdot 4}{64 \cdot 63}$.

Alternatively : $n(A) = 7 \cdot 8 + 7 \cdot 8 = 112$

Ask: Prob that they have a corner in common]

Q.14₁₈₂ Two red counters, three green counters and 4 blue counters are placed in a row in random order. The probability that no two blue counters are adjacent is

- (A) $\frac{7}{99}$ (B) $\frac{7}{198}$ (C*) $\frac{5}{42}$ (D) none

[Sol. R R G G G B B B B when counters are alike

$$n(S) = \frac{9!}{2!3!4!}$$

$$n(A) = \frac{5!}{3!2!} \cdot {}^6C_4 \quad | R | R | G | G | G |$$

$$\therefore P(A) = \frac{5! \cdot 15}{3!2! \cdot 9!} = \frac{2!3!4!}{9!} = \frac{6! \cdot 60}{9 \cdot 8 \cdot 7 \cdot 6!} = \frac{60}{7 \cdot 8 \cdot 9} = \frac{15}{7 \cdot 2 \cdot 9} = \frac{5}{42}$$

Alternatively : $n(S) = 9!$ $R_1 R_1 G_1 G_2 G_3 B_1 B_2 B_3 B_4$
 $n(A) = 5! \cdot {}^6C_4 \cdot 4!$ when counters are different

$$p = \frac{5! \cdot 6 \cdot 5 \cdot 4 \cdot 3}{9!} = \frac{5 \cdot 4 \cdot 3}{9 \cdot 8 \cdot 7} = \frac{5}{42}]$$

Q.15 The probabilities that a student will receive A, B, C or D grade are 0.40, 0.35, 0.15 and 0.10 respectively. Find the probability that a student will receive

- (i) not an A grade (ii) B or C grade (iii) at most C grade
 [Ans. (i) 0.6, (ii) 0.5, (iii) 0.25]

Q.16 In a single throw of three dice, determine the probability of getting

- (i) a total of 5 (ii) a total of at most 5 (iii) a total of at least 5.

$$[Ans. (i) \frac{1}{36}, (ii) \frac{5}{108}, (iii) \frac{53}{54}]$$

Q.17 A die is thrown once. If E is the event "the number appearing is a multiple of 3" and F is the event "the number appearing is even", find the probability of the event "E and F". Are the events E and F independent?

$$[Ans. P(E) = \frac{1}{3}, P(F) = \frac{1}{2}, P(E \text{ and } F) = \frac{1}{6}; \text{Yes}]$$

Q.18 In the two dice experiment, if E is the event of getting the sum of number on dice as 11 and F is the event of getting a number other than 5 on the first die, find P(E and F). Are E and F independent events?

$$[Ans. P(E) = \frac{2}{36}, P(F) = \frac{30}{36}, P(E \cap F) = \frac{1}{36}; \text{Not independent}]$$

Q.19 A natural number x is randomly selected from the set of first 100 natural numbers. Find the probability

that it satisfies the inequality. $x + \frac{100}{x} > 50$ [Ans: $\frac{55}{100} = \frac{11}{20}$]

Note: $\{1, 2, 48, 49, 50, \dots, 100\}$ [wrong Ans given by students $\frac{1}{50}, \frac{27}{50}, \frac{53}{100}$]

- Q.20 3 students A and B and C are in a swimming race. A and B have the same probability of winning and each is twice as likely to win as C. Find the probability that B or C wins. Assume no two reach the winning point simultaneously.

[Sol. $P(C) = p$; $P(A) = 2p$; $P(B) = 2p$
 $\therefore 5p = 1 \Rightarrow p = 1/5$

$$P(B \text{ or } C) = P(B) + P(C) = \frac{2}{5} + \frac{1}{5} = \frac{3}{5} \quad]$$

- Q.21 A box contains 7 tickets, numbered from 1 to 7 inclusive. If 3 tickets are drawn from the box, one at a time, determine the probability that they are alternatively either odd-even-odd or even-odd-even.

$$[\text{Ans. } p = \frac{4 \cdot 3 \cdot 3 + 3 \cdot 4 \cdot 2}{7 \cdot 6 \cdot 5} = \frac{6}{210} = \frac{2}{7} \quad]$$

- Q.22 5 different marbles are placed in 5 different boxes randomly. Find the probability that exactly two boxes remain empty. Given each box can hold any number of marbles.

[Sol. $n(S) = 5^5$; For computing favourable outcomes.
 2 boxes which are to remain empty, can be selected in 5C_2 ways and 5 marbles can be placed in the remaining 3 boxes in groups of 221 or 311 in

$$3! \left[\frac{5!}{2!2!2!} + \frac{5!}{3!2!} \right] = 150 \text{ ways} \Rightarrow n(A) = {}^5C_2 \cdot 150$$

$$\text{Hence } P(E) = {}^5C_2 \cdot \frac{150}{5^5} = \frac{60}{125} = \frac{12}{25} \quad \text{Ans.}]$$

- Q.23₁₃₂ South African cricket captain lost the toss of a coin 13 times out of 14. The chance of this happening was

$$(A^*) \frac{7}{2^{13}} \quad (B) \frac{1}{2^{13}} \quad (C) \frac{13}{2^{14}} \quad (D) \frac{13}{2^{13}}$$

[Hint: L and W can be filled at 14 places in 2^{14} ways.

$$\therefore n(S) = 2^{14}.$$

Now 13 L's and 1 W can be arranged at 14 places in 14 ways.

$$\text{Hence } n(A) = 14$$

$$\therefore p = \frac{14}{2^{14}} = \frac{7}{2^{13}} \quad]$$

- Q.24 There are ten prizes, five A's, three B's and two C's, placed in identical sealed envelopes for the top ten contestants in a mathematics contest. The prizes are awarded by allowing winners to select an envelope at random from those remaining. When the 8th contestant goes to select the prize, the probability that the remaining three prizes are one A, one B and one C, is

$$(A^*) 1/4 \quad (B) 1/3 \quad (C) 1/12 \quad (D) 1/10$$

[Hint: $n(S) = {}^{10}C_7 = 120$

$$n(A) = {}^5C_4 \cdot {}^3C_2 \cdot {}^2C_1$$

$$P(E) = \frac{5 \cdot 3 \cdot 2}{120} = \frac{1}{4} \quad \text{Ans.}]$$